Assignment-12.3

HT.NO-2403A52L06

BATCH-50

Q: Lab 12: Algorithms with AI Assistance Sorting, Searching, and Algorithm Optimization Using AI Tools

Lab Objectives

The objectives of this laboratory exercise are to:

Week3 - Wednesday

- Apply AI-assisted programming techniques to implement sorting and searching algorithms.
- Analyze and compare algorithm efficiency using time and space complexity.
- Understand how AI tools can suggest optimizations and alternative algorithmic approaches.
- Strengthen problem-solving skills through real-world, data-driven scenarios.

Learning Outcomes

After completing this lab, students will be able to:

- Implement and optimize classic algorithms using AI-assisted coding tools.
- Compare multiple algorithms for the same problem and justify their selection.
- Measure and analyze runtime performance using experimental data.
- Critically review and refine AI-generated algorithmic solutions.

Task 1: Sorting Student Records for Placement Drive

Scenario

SR University's Training and Placement Cell needs to shortlist candidates efficiently during campus placements. Student records must be sorted by CGPA in descending order.

Tasks

1. Use GitHub Copilot to generate a program that stores student records (Name, Roll Number, CGPA).
2. Implement the following sorting algorithms using AI assistance:

o Quick Sort o

Merge Sort

3. Measure and compare runtime performance for large datasets.
4. Write a function to display the top 10 students based on CGPA.

Expected Outcome

• Correctly sorted student records.

• Performance comparison between Quick Sort and Merge Sort.

• Clear output of top-performing students.

Task 2: Implementing Bubble Sort with AI Comments •

Task: Write a Python implementation of Bubble Sort.

• Instructions:

• Students implement Bubble Sort normally.

• Ask AI to generate inline comments explaining key logic (like swapping, passes, and termination).

• Request AI to provide time complexity analysis.

• Expected Output:

• A Bubble Sort implementation with AI-generated explanatory comments and complexity analysis.

Task 3: Quick Sort and Merge Sort Comparison

• Task: Implement Quick Sort and Merge Sort using recursion.

• Instructions:

• Provide AI with partially completed functions for recursion.

• Ask AI to complete the missing logic and add docstrings.

• Compare both algorithms on random, sorted, and reverse-sorted lists.

• Expected Output:

• Working Quick Sort and Merge Sort implementations.

• AI-generated explanation of average, best, and worst-case complexities.

Task 4 (Real-Time Application – Inventory Management System)

Scenario: A retail store's inventory system contains thousands of products, each with attributes like product ID, name, price, and stock quantity. Store staff need to:

1. Quickly search for a product by ID or name.
2. Sort products by price or quantity for stock analysis.

Task:

• Use AI to suggest the most efficient search and sort algorithms for this use case.

• Implement the recommended algorithms in Python.

• Justify the choice based on dataset size, update frequency, and performance requirements.

Expected Output:

• A table mapping operation → recommended algorithm → justification.

• Working Python functions for searching and sorting the inventory.

Task 5: Real-Time Stock Data Sorting & Searching Scenario:

An AI-powered FinTech Lab at SR University is building a tool for analyzing stock price movements. The requirement is to quickly sort stocks by daily gain/loss and search for specific stock symbols efficiently.

• Use GitHub Copilot to fetch or simulate stock price data (Stock Symbol, Opening Price, Closing Price).

• Implement sorting algorithms to rank stocks by percentage change.

• Implement a search function that retrieves stock data instantly when a stock symbol is entered.

• Optimize sorting with Heap Sort and searching with Hash Maps.

• Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

CODE:

```
import time
import random
from typing import List, Tuple, Dict
from collections import defaultdict
import heapq

# ==================== TASK 1: Student Records Sorting ====================

class StudentRecord:
    """Class to store student information"""
    def __init__(self, name: str, roll_number: str, cgpa: float):
        self.name = name
        self.roll_number = roll_number
        self.cgpa = cgpa

    def __repr__(self):
```

```
        return f"Student(Name: {self.name}, Roll: {self.roll_number}, CGPA: {self.cgpa})"

    def __lt__(self, other):
        return self.cgpa > other.cgpa  # For descending order

# Quick Sort Implementation for Student Records
def quick_sort_students(records: List[StudentRecord], low: int = 0, high: int = None) -> List[StudentRecord]:
    """
    Quick Sort algorithm for sorting student records by CGPA in descending order.

    Time Complexity: O(n log n) average, O(n²) worst case
    Space Complexity: O(log n) due to recursion

    Args:
        records: List of StudentRecord objects
```

```
        low: Start index
        high: End index

    Returns:
        Sorted list of student records
    """
    if high is None:
        high = len(records) - 1

    if low < high:
        # Partition and get pivot index
                                    pi        =
_partition_students(records,   low,
high)
        # Recursively sort left and right
parts
         quick_sort_students(records,
low, pi - 1)
         quick_sort_students(records,
pi + 1, high)

    return records

def    _partition_students(records:
List[StudentRecord], low: int, high:
int) -> int:
     """Helper function to partition
the list"""
```

```
    pivot = records[high].cgpa
    i = low - 1

    for j in range(low, high):
        if records[j].cgpa > pivot:  # Descending order
            i += 1
                records[i], records[j] = records[j], records[i]

    records[i + 1], records[high] = records[high], records[i + 1]
    return i + 1

# Merge Sort Implementation for Student Records
def merge_sort_students(records: List[StudentRecord]) -> List[StudentRecord]:
    """
    Merge Sort algorithm for sorting student records by CGPA in descending order.

    Time Complexity: O(n log n) in all cases
        Space Complexity: O(n) for temporary arrays
```

```
    Args:
        records: List of StudentRecord
objects

    Returns:
        Sorted list of student records
    """
    if len(records) <= 1:
        return records

    mid = len(records) // 2
                                left              =
merge_sort_students(records[:mi
d])
                               right              =
merge_sort_students(records[mid
:])

        return   _merge_students(left,
right)

def              _merge_students(left:
List[StudentRecord],              right:
List[StudentRecord])                  ->
List[StudentRecord]:
    """Helper function to merge two
sorted lists"""
```

```
    result = []
    i = j = 0

        while i < len(left) and j <
len(right):
                if left[i].cgpa >=
right[j].cgpa:  # Descending order
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

def
display_top_10_students(records:
List[StudentRecord]) -> None:
      """Display top 10 students by
CGPA"""
    print("\n" + "="*60)
       print("TOP 10 STUDENTS BY
CGPA")
    print("="*60)
     print(f"{'Rank':<6}{'Name':<20}{
'Roll Number':<15}{'CGPA':<10}")
```

```
    print("-"*60)

    for i, student in enumerate(records[:10], 1):
        print(f"{i:<6}{student.name:<20}{student.roll_number:<15}{student.cgpa:<10.2f}")
    print("="*60)

# ==================== TASK 2: Bubble Sort with Comments ====================

def bubble_sort(arr: List[float]) -> List[float]:
    """
    Bubble Sort Algorithm - Simplest sorting algorithm

    Time Complexity Analysis:
    - Best Case: O(n) - when array is already sorted
    - Average Case: O(n²) - random distribution
    - Worst Case: O(n²) - when array is reverse sorted
```

```
    Space Complexity: O(1) - sorts in-place

    Args:
        arr: List of numbers to be sorted

    Returns:
        Sorted list in ascending order
    """
    n = len(arr)

    # Outer loop: controls number of passes
    # Each pass moves the largest unsorted element to its correct position
    for i in range(n):
        # Flag to optimize: if no swaps occur, array is sorted
        swapped = False

        # Inner loop: compares adjacent elements
        # i elements at the end are already in correct position, so we reduce range
        for j in range(0, n - i - 1):
```

```
            # Comparison: swap if current element is greater than next element
            if arr[j] > arr[j + 1]:
                # Swap operation: exchange positions of two elements
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True

        # Optimization: if no swaps occurred in a pass, array is sorted
        # Termination condition: exit early if array becomes sorted
        if not swapped:
            break

    return arr

# ==================== TASK 3: Quick Sort vs Merge Sort ====================

def quick_sort_generic(arr: List[float], low: int = 0, high: int = None) -> List[float]:
    """
```

Quick Sort Implementation using Recursion

Complexity Analysis:
- Best Case: O(n log n) - balanced partition
- Average Case: O(n log n)
- Worst Case: O(n^2) - pivot is always smallest/largest

Space Complexity: O(log n) - recursion stack

Args:
arr: List to sort
low: Starting index
high: Ending index

Returns:
Sorted list
"""
if high is None:
high = len(arr) - 1

if low < high:
pi = _partition_generic(arr, low, high)

```
        quick_sort_generic(arr, low, pi
- 1)
        quick_sort_generic(arr, pi + 1,
high)

    return arr

def             _partition_generic(arr:
List[float], low: int, high: int) -> int:
    """
    Partition helper for Quick Sort
    Selects pivot and partitions array
into smaller and larger elements
    """
    pivot = arr[high]
    i = low - 1

    for j in range(low, high):
        if arr[j] < pivot:
            i += 1
            arr[i], arr[j] = arr[j], arr[i]

    arr[i + 1], arr[high] = arr[high],
arr[i + 1]
    return i + 1

def          merge_sort_generic(arr:
List[float]) -> List[float]:
```

```
    """
        Merge Sort Implementation using Recursion

    Complexity Analysis:
    - Best Case: O(n log n)
    - Average Case: O(n log n)
    - Worst Case: O(n log n)

        Space Complexity: O(n) - requires temporary arrays for merging

    Args:
        arr: List to sort

    Returns:
        Sorted list
    """
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
                left = merge_sort_generic(arr[:mid])
                right = merge_sort_generic(arr[mid:])
```

```
        return _merge_generic(left,
right)

def _merge_generic(left:
List[float], right: List[float]) ->
List[float]:
    """
    Merge helper for Merge Sort
    Combines two sorted arrays into
one sorted array
    """
    result = []
    i = j = 0

        while i < len(left) and j <
len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result
```

```
# ==================== TASK 4: Inventory Management System ====================

class Product:
    """Product class for inventory management"""
  def __init__(self, product_id: int, name: str, price: float, quantity: int):
      self.product_id = product_id
      self.name = name
      self.price = price
      self.quantity = quantity

  def __repr__(self):
              return f"Product(ID: {self.product_id}, Name: {self.name}, Price: ${self.price}, Qty: {self.quantity})"

class InventorySystem:
        """Inventory Management System with optimized search and sort"""

  def __init__(self):
     self.products: List[Product] = []
```

```
        self.id_index: Dict[int, Product] = {}  # Hash map for O(1) ID lookup
        self.name_index: Dict[str, List[Product]] = defaultdict(list)  # For name search

    def add_product(self, product: Product):
        """Add product to inventory"""
        self.products.append(product)
        self.id_index[product.product_id] = product
        self.name_index[product.name.lower()].append(product)

    def search_by_id(self, product_id: int) -> Product:
        """
        Search product by ID - O(1) using hash map
        Recommended: Hash Map for exact ID match
        """
        return self.id_index.get(product_id, None)
```

```
    def search_by_name(self, name:
str) -> List[Product]:
        """
        Search product by name - O(1)
average using hash map
        Recommended: Hash Map for
name lookups
        """
                                return
self.name_index.get(name.lower()
, [])

        def sort_by_price(self) ->
List[Product]:
        """
        Sort products by price - O(n log
n) using Merge Sort
        Recommended: Merge Sort for
consistent O(n log n)
        """
                                return
merge_sort_generic([p.price for p
in self.products])

        def sort_by_quantity(self) ->
List[Product]:
        """
```

Sort products by quantity - O(n log n)

Recommended: Quick Sort for average case efficiency

"""

return quick_sort_generic([p.quantity for p in self.products])

========== TASK 5: Stock Data Analysis ==========

class StockData:

"""Class to store stock information"""

def __init__(self, symbol: str, opening_price: float, closing_price: float):

self.symbol = symbol

self.opening_price = opening_price

self.closing_price = closing_price

self.daily_change = closing_price - opening_price

self.percentage_change = (self.daily_change /

```
opening_price) * 100 if
opening_price != 0 else 0

    def __repr__(self):
        return f"Stock({self.symbol},
Open: ${self.opening_price:.2f},
Close: ${self.closing_price:.2f},
Change:
{self.percentage_change:.2f}%)"

def generate_stock_data(count:
int = 100) -> List[StockData]:
    """
    Simulate stock price data for
testing

    Args:
        count: Number of stocks to
generate

    Returns:
        List of StockData objects
    """
    symbols = ['AAPL', 'GOOGL',
'MSFT', 'AMZN', 'TSLA', 'META',
'NVDA', 'JPM', 'BA', 'WMT']
    stocks = []
```

```
    for i in range(count):
                symbol = random.choice(symbols) + str(i % 10)
                opening = round(random.uniform(50, 500), 2)
            closing = round(opening * random.uniform(0.95, 1.05), 2)
        stocks.append(StockData(symbol, opening, closing))

    return stocks

class StockAnalysisSystem:
    """Stock data analysis system with optimized search and sort"""

    def __init__(self, stocks: List[StockData]):
        self.stocks = stocks
        self.symbol_map: Dict[str, StockData] = {stock.symbol: stock for stock in stocks}

    def search_by_symbol(self, symbol: str) -> StockData:
        """
```

Search stock by symbol using Hash Map - O(1)

Recommended: Hash Map for instant lookup

"""

return self.symbol_map.get(symbol, None)

def sort_by_percentage_gain(self) -> List[StockData]:

"""

Sort stocks by percentage change using Heap Sort - O(n log n)

Recommended: Heap Sort for consistent performance

"""

return sorted(self.stocks, key=lambda x: x.percentage_change, reverse=True)

def sort_by_gain_heap_sort(self) -> List[StockData]:

"""

```
        Heap Sort implementation for stock sorting

        Time Complexity: O(n log n) guaranteed
        Space Complexity: O(1) if not counting output
        """
        heap = []
        for stock in self.stocks:
            # negate for max heap behavior
            heapq.heappush(heap, (-stock.percentage_change, stock.symbol, stock))

        result = []
        while heap:
            _, _, stock = heapq.heappop(heap)
            result.append(stock)

        return result

    def get_top_gainers(self, n: int = 10) -> List[StockData]:
        """Get top n stocks by percentage gain"""
```

```
        return sorted(self.stocks, key=lambda x: x.percentage_change, reverse=True)[:n]

    def get_top_losers(self, n: int = 10) -> List[StockData]:
        """Get top n stocks by percentage loss"""
        return sorted(self.stocks, key=lambda x: x.percentage_change)[:n]

# ==================== PERFORMANCE TESTING ====================

def measure_sorting_performance(data: List[float], algorithm_name: str, algorithm_func) -> Tuple[float, List[float]]:
    """
        Measure sorting algorithm performance

    Returns:
```

```
        Tuple of (execution_time, sorted_data)
    """
    start_time = time.time()
                sorted_data = algorithm_func(data.copy())
    end_time = time.time()

    return end_time - start_time, sorted_data

def performance_comparison():
    """Compare Quick Sort and Merge Sort on different datasets"""
    sizes = [100, 500, 1000]

    print("\n" + "="*70)
                print("PERFORMANCE COMPARISON: QUICK SORT vs MERGE SORT")
    print("="*70)
                    print(f"{'Dataset Size':<20}{'Quick Sort (sec)':<20}{'Merge Sort (sec)':<20}{'Winner':<10}")
    print("-"*70)
```

```
    for size in sizes:
        # Random data
            data = [random.uniform(1,
1000) for _ in range(size)]

                    qs_time, _ =
measure_sorting_performance(da
ta, "Quick Sort",
quick_sort_generic)
                    ms_time, _ =
measure_sorting_performance(da
ta, "Merge Sort",
merge_sort_generic)

            winner = "Quick Sort" if
qs_time < ms_time else "Merge
Sort"
        print(f"{size:<20}{qs_time:<2
0.6f}{ms_time:<20.6f}{winner:<10
}")

    print("="*70)

# ==================== MAIN
EXECUTION
====================

if __name__ == "__main__":
```

```
    print("\n" + "="*70)
        print("LAB 12: SORTING, SEARCHING, AND ALGORITHM OPTIMIZATION")
    print("="*70)

    # TASK 1: Student Records
      print("\n--- TASK 1: Student Records Sorting ---")
    students = [
                StudentRecord("Akhir Ahmed", "001", 9.2),
       StudentRecord("Priya Singh", "002", 8.9),
          StudentRecord("Raj Patel", "003", 9.5),
        StudentRecord("Sara Khan", "004", 8.7),
      StudentRecord("Amit Kumar", "005", 9.1),
        StudentRecord("Lisa Wong", "006", 9.3),
       StudentRecord("John Smith", "007", 8.5),
      StudentRecord("Emma Davis", "008", 9.4),
                StudentRecord("Rahul Verma", "009", 8.8),
```

```
        StudentRecord("Zoe Chen",
"010", 9.6),
    ]

    # Quick Sort
                students_qs =
[StudentRecord(s.name,
s.roll_number, s.cgpa) for s in
students]
    qs_start = time.time()
    quick_sort_students(students_q
s)
    qs_time = time.time() - qs_start

    # Merge Sort
                students_ms =
[StudentRecord(s.name,
s.roll_number, s.cgpa) for s in
students]
    ms_start = time.time()
    merge_sort_students(students_
ms)
    ms_time = time.time() - ms_start

        print(f"Quick Sort Time:
{qs_time:.6f} seconds")
        print(f"Merge Sort Time:
{ms_time:.6f} seconds")
```

```
    display_top_10_students(students_qs)

    # TASK 2: Bubble Sort
    print("\n--- TASK 2: Bubble Sort with Comments ---")
    test_array = [64, 34, 25, 12, 22, 11, 90]
    print(f"Original Array: {test_array}")
    sorted_array = bubble_sort(test_array.copy())
    print(f"Sorted Array: {sorted_array}")
    print("Time Complexity: O(n²) average and worst case, O(n) best case")
    print("Space Complexity: O(1)")

    # TASK 3: Quick vs Merge Sort Comparison
    print("\n--- TASK 3: Quick Sort vs Merge Sort Comparison ---")
    performance_comparison()

    # TASK 4: Inventory Management
```

```
    print("\n--- TASK 4: Inventory Management System ---")
    inventory = InventorySystem()
    inventory_products = [
        Product(1, "Laptop", 999.99, 50),
        Product(2, "Mouse", 29.99, 200),
        Product(3, "Keyboard", 79.99, 150),
        Product(4, "Monitor", 299.99, 75),
        Product(5, "Headphones", 149.99, 100),
    ]

    for product in inventory_products:
        inventory.add_product(product)

    print("Inventory Management - Algorithm Recommendations:")
    print("="*60)
    print("Operation\t\tRecommended Algorithm\tJustification")
    print("-"*60)
```

```
    print("Search by ID\t\tHash Map\t\tO(1) lookup")
    print("Search by Name\t\tHash Map\t\tO(1) average lookup")
    print("Sort by Price\t\tMerge Sort\t\tO(n log n) guaranteed")
    print("Sort by Quantity\tQuick Sort\t\tO(n log n) average")
    print("="*60)

    # TASK 5: Stock Analysis
    print("\n--- TASK 5: Stock Data Sorting & Searching ---")
    stocks = generate_stock_data(20)
    stock_system = StockAnalysisSystem(stocks)

    print("\nTop 5 Gainers:")
    print("-"*70)
    for i, stock in enumerate(stock_system.get_top_gainers(5), 1):
        print(f"{i}. {stock.symbol}: {stock.percentage_change:+.2f}%")

    print("\nTop 5 Losers:")
```

```
    print("-"*70)
    for i, stock in enumerate(stock_system.get_top_losers(5), 1):
        print(f"{i}. {stock.symbol}: {stock.percentage_change:+.2f}%")

    # Search example
    search_symbol = stocks[0].symbol
    found_stock = stock_system.search_by_symbol(search_symbol)
    print(f"\nSearch Result for {search_symbol}: {found_stock}")

    print("\n" + "="*70)
    print("="*70)
```

OUTPUT:

```
PS C:\Users\akhir\OneDrive\Documents\AI ASSISTANT> & C:\Users\akhir\AppData\Local\Python\pythoncore-3.14-64\python.exe
"


======================================================================
LAB 12: SORTING, SEARCHING, AND ALGORITHM OPTIMIZATION
======================================================================

--- TASK 1: Student Records Sorting ---
Quick Sort Time: 0.000011 seconds
Merge Sort Time: 0.000020 seconds

============================================================
TOP 10 STUDENTS BY CGPA
============================================================
Rank  Name                Roll Number    CGPA
------------------------------------------------------------
1     Zoe Chen            010            9.60
2     Raj Patel           003            9.50
3     Emma Davis          008            9.40
4     Lisa Wong           006            9.30
5     Akhir Ahmed         001            9.20
6     Amit Kumar          005            9.10
7     Priya Singh         002            8.90
8     Rahul Verma         009            8.80
9     Sara Khan           004            8.70
10    John Smith          007            8.50
============================================================

--- TASK 2: Bubble Sort with Comments ---
Original Array: [64, 34, 25, 12, 22, 11, 90]
Sorted Array: [11, 12, 22, 25, 34, 64, 90]
Time Complexity: O(n²) average and worst case, O(n) best case
Space Complexity: O(1)
```

```
--- TASK 3: Quick Sort vs Merge Sort Comparison ---

======================================================================
PERFORMANCE COMPARISON: QUICK SORT vs MERGE SORT
======================================================================
Dataset Size        Quick Sort (sec)    Merge Sort (sec)    Winner
----------------------------------------------------------------------
100                 0.000055            0.000117            Quick Sort
500                 0.000295            0.000588            Quick Sort
1000                0.000788            0.001117            Quick Sort
======================================================================

--- TASK 4: Inventory Management System ---
Inventory Management - Algorithm Recommendations:
============================================================
Operation               Recommended Algorithm   Justification
------------------------------------------------------------
Search by ID            Hash Map                O(1) lookup
Search by Name          Hash Map                O(1) average lookup
Sort by Price           Merge Sort              O(n log n) guaranteed
Sort by Quantity        Quick Sort              O(n log n) average
============================================================
```

```
--- TASK 5: Stock Data Sorting & Searching ---

Top 5 Gainers:
----------------------------------------------------------------------
1. BA2: +3.79%
2. GOOGL3: +3.16%
3. JPM8: +3.00%
4. BA0: +2.87%
5. META6: +2.78%

Top 5 Losers:
----------------------------------------------------------------------
1. GOOGL5: -4.55%
2. NVDA6: -4.52%
3. NVDA4: -4.42%
4. WMT0: -3.00%
5. META7: -2.95%

Search Result for BA0: Stock(BA0, Open: $387.73, Close: $398.85, Change: 2.87%)

======================================================================
======================================================================
PS C:\Users\akhir\OneDrive\Documents\AI ASSISTANT>
```